

Real-Time Path Planning for Tethered Robots in Extreme Environments Based on Constraint Consistency and Optimality

Abstract—The tether enables the robot to operate in extreme environments while imposing constraints on length and the maximum deployment angle in path planning. Conventional path-planning algorithms for tethered robots suffer from tight coupling between constraint evaluation and the search process, resulting in low planning efficiency. To address this issue, we propose a method based on our constraint consistency and optimality (CCO) theorem. Instead of computing tether constraints during the search for robot paths, we first search for terminal tether configurations and evaluate the constraints only afterward, effectively decoupling constraint evaluation from the search process. In addition, we introduce the Receptive Field Enhanced Dijkstra (RFE-Dijkstra) algorithm, which significantly reduces the search space and further accelerates computation. Experiments show that the proposed CCO-based planner runs in milliseconds on grid maps with dozens of obstacles, using only 0.17% of the computation time of direct shortest-path search and just 0.014% of that of the method that precomputes the reachable space before searching.

Index Terms—tethered robot, constrained path planning, constraint consistency and optimality.

I. INTRODUCTION

TETHERED robots are a class of mobile robots connected to a base station through a tether, as illustrated in Fig. 1. The tether provides attitude stabilization, communication support, and continuous power supply, enabling robots to perform high-risk tasks such as cliff sampling [1]–[3], mine rescue [4], [5], and seabed exploration [6], [7] in extreme environments. However, the presence of the tether introduces additional constraints into the path-planning problem. Due to the limited tether length, geometric length constraints must be considered [5], [8]. Moreover, when operating on steep slopes, the robot relies on tether tension to maintain force balance [9], [10]. From a mechanical perspective, the tension must lie within a certain angular range to counteract gravity, leading to a maximum deployment angle constraint.

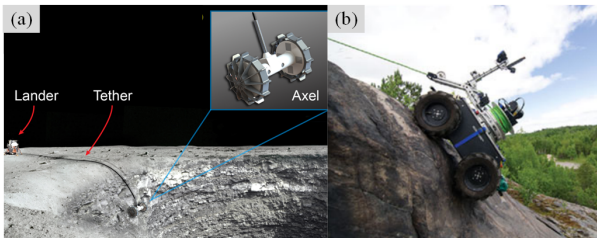


Fig. 1. Tethered robots working in extreme environments (a) Axel [11] is designed by NASA to rappel into steep planetary craters and lava-tube skylights. (b) TRex [12] is designed for 3D mapping of steep terrain and harsh environments.

Unlike conventional kinematic constraints that depend solely on robot configurations, tether-related constraints arise from interactions between the tether and environmental obstacles, i.e., they depend on the tether configuration. Most existing approaches address path planning from the robot configuration perspective, which requires computing the tether configuration at each expanded node during the search process [13]. This tight coupling between node expansion and constraint evaluation leads to significant computational overhead and limits real-time performance.

To address this issue, this paper proposes a novel theoretical framework that fully decouples constraint evaluation from the search process. By jointly considering the mechanical and topological properties of a taut tether, we establish a triangular relationship among the initial tether configuration, the terminal tether configuration, and the robot path. Based on this relationship, we derive the Constraint Consistency and Optimality (CCO) theorem, which transforms the multi-constraint path-planning problem into selecting feasible terminal tether configurations.

Based on the CCO theorem, we develop a path-planning framework that directly searches terminal tether configurations instead of robot paths. To improve efficiency, we propose a receptive-field-enhanced Dijkstra algorithm (RFE-Dijkstra) to prune infeasible configurations without explicit tether computation, and an h-signature-enhanced curve-shortening algorithm (HSE-CS) to address the limitation of standard curve-shortening (CS) method [14] in accurately computing tether configurations under the maximum deployment angle constraint. The main contributions of this work are summarized as follows:

- 1) We establish the Constraint Consistency and Optimality (CCO) theorem, providing a theoretical foundation for multi-constraint path planning of tethered robots.
- 2) We propose a CCO-based planning framework, along with the RFE-Dijkstra and HSE-CS algorithms, to efficiently handle tether length and maximum deployment angle constraints.
- 3) We validate the proposed method in environments with varying obstacle densities, demonstrating both correctness and significant improvements in computational efficiency.

II. RELATED WORK

The goal of tethered robot path planning is to search for feasible robot paths satisfying tether constraints within a complex homotopy class space. Early work [15] used geometric methods to classify homotopy classes in 2D spaces, but these are not well-suited for direct path search. Subsequently,

Bhattacharya *et al.* introduced the h-signature to distinguish trajectories, enabling graph-search-based planners to compute minimum-cost paths under tether constraints [16]. However, due to the large-scale topology-augmented planning space and the strong coupling between constraint evaluation and search, computational complexity remains high. Accordingly, existing methods for reducing computational complexity in tethered robot path planning can be broadly grouped into three categories.

The first category consists of map preprocessing-based methods. Classical approaches [17]–[19] model the environment as sparse visibility graphs, reducing search-space dimensionality and improving efficiency, but their strong geometric assumptions limit applicability in complex environments. Kim *et al.* constructed a homotopy-augmented graph based on h-signatures and proposed a preprocessing-based method to compute the reachable space for a fixed anchor [14]. By shifting computation from online planning to offline preprocessing, efficiency is improved; however, the reachable space must be recomputed when the anchor changes.

The second category consists of heuristic search-based methods. Kim *et al.* proposed a Multi-Heuristic A* framework with topology-based heuristics [20], reducing computational complexity by decreasing node expansions, but yielding generally suboptimal solutions. Kumar *et al.* uses admissible heuristics to search shortest paths on sparse tangential graphs [2]; however, constraint evaluation remains tightly coupled with search, failing to fundamentally address computational complexity.

More recently, there has been a study attempting to reduce computational complexity from a theoretical perspective by decoupling constraint evaluation from search. Yang *et al.* proved the consistency of the tether-length constraint, enabling its decoupling from the search process [21]. However, this study places excessive emphasis on the topological characteristics of the tether, neglects the mechanical properties of a taut tether, and lacks discussion of the tether deformation process during robot motion. These limitations restrict the theorem to tether-length constraints, making it difficult to extend to other tether-related constraints, such as the maximum deployment angle.

III. THEORETICAL FOUNDATION

A. Problem Statement

In this work, the planning problem is formulated as a two-dimensional path planning problem, as illustrated in Fig. 2(a). The modeling is based on the physical observation that, under the combined effects of gravity and anchor tension, the tether remains taut and closely adheres to the terrain surface, thereby constraining its motion to a two-dimensional manifold embedded in three-dimensional space. Based on this observation, the terrain is approximated as a planar slope with a constant inclination angle β , which can be viewed as a global planar parameterization of the underlying manifold. This approximation is reasonable for terrains with slowly varying geometry or locally uniform slope and has been widely adopted in prior works under similar assumptions [10], [22].

TABLE I
NOTATION AND PARAMETERS USED IN THIS PAPER

Symbol	Description
M	The entire planning map
M_{free}	The free space within the planning map
M_{obs}	The obstacle region within the planning map
β	Slope angle
μ	Equivalent coefficient of friction between the tethered robot and the slope
$\alpha_*, \alpha(s)$	Tether configuration or curve
$\alpha^-, \alpha^-(s)$	The reverse curve of α
α_s, α_g	Initial tether configuration, terminal tether configuration
α_{sg}	The shortest path in the homotopy class
α_{sg}^*	The globally shortest path satisfying the tether constraints
$[\alpha_{s \rightarrow g}]$	The set of all tether configurations along α_{sg}
p_b, p_s, p_g	Anchor point, start point, goal point
$L(\alpha)$	The length of α
$l, \bar{l}$	Tether length, maximum length
$\Gamma(\alpha)$	The maximum deployment angle of α
$\gamma, \bar{\gamma}$	Tether deployment angle, maximum deployment angle

The symbol descriptions used in this paper are provided in Tab. I. The planar terrain is denoted as M , with M_{free} and M_{obs} representing the free space and obstacle region, respectively. We assume that the planning algorithm operates in an offline manner and that the locations of all obstacles are known a priori. The projection of the anchor onto M is denoted as p_b , and the robot's current position is denoted as p_s , with the corresponding tether configuration α_s . The robot's goal position is denoted as p_g , and the planned path is denoted as α_{sg} . As the robot moves along α_{sg} toward p_g , the tether configuration evolves to α_g . The set of all intermediate tether configurations along this motion is denoted by $[\alpha_{s \rightarrow g}]$.

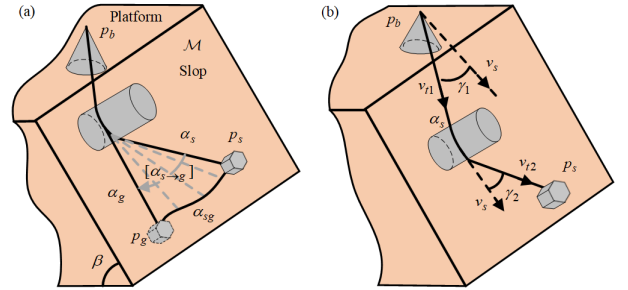


Fig. 2. Illustration of path planning problem for tethered robots on the slope. (a) Tether deformation as the robot moves from the start position p_s to the goal p_g on a slope. (b) Maximum deployment angle of the tether. The angle between v_s and v_t represents the deployment angle.

Path planning for a tethered robot on a slope must satisfy not only the tether length constraint but also the maximum deployment angle constraint to prevent slippage. Let μ denote the equivalent friction coefficient between the robot and the slope. In this study, we consider the case $\mu < \tan \beta$, i.e., the robot requires the assistance of the tether to remain stable on the slope. To avoid slippage, the system becomes unstable if the deployment angle exceeds 90° , which defines the feasible range as $\alpha \in [0, \frac{\pi}{2}]$. Within this range, the maximum deployment angle constraint is jointly determined by the robot configuration and environmental factors, specifically the friction coefficient and the slope angle, the maximum deployment angle is $\bar{\gamma} = \arcsin(\mu \cot \beta)$.

For an arbitrary tether configuration, let $\mathbf{v}_t$ denote the unit vector along the tether and $\mathbf{v}_s$ the unit vector along the downslope direction, as illustrated in Fig. 2(b). The maximum deployment angle of the configuration, α , is then defined as

$$\Gamma(\alpha) = \max_{1 \leq i \leq n} \arccos(\langle \mathbf{v}_{ti}, \mathbf{v}_s \rangle), \quad (1)$$

where $\langle \cdot, \cdot \rangle$ denotes the inner product.

In this planning problem, the tether is subject to a maximum length $\bar{l}$ and a maximum deployment angle $\bar{\gamma}$. The objective of the planning problem is to plan a globally shortest path α_{sg}^* such that all tether configurations along the path, $[\alpha_{s \rightarrow g}]$, satisfy

$$L(\alpha_i) \leq \bar{l}, \quad \Gamma(\alpha_i) \leq \bar{\gamma}, \quad \forall \alpha_i \in [\alpha_{s \rightarrow g}].$$

B. Definition and Lemma

Definition 1 (Curve, Homotopy, and Homotopy Class) A curve α is defined as a continuous mapping from the unit interval to the free space M_{free} :

$$\alpha : [0, 1] \rightarrow M_{\text{free}}, \quad s \mapsto \alpha(s), \quad (2)$$

where $\alpha(0)$ and $\alpha(1)$ denote the start and end points, respectively. The reverse of a curve is defined as

$$\alpha^-(s) := \alpha(1 - s), \quad s \in [0, 1]. \quad (3)$$

If two curves α_1 and α_2 satisfy $\alpha_1(1) = \alpha_2(0)$, their concatenation is denoted as $\alpha_1 * \alpha_2$. Two curves α_1, α_2 with the same start and goal points are said to be *homotopic*, denoted as $\alpha_1 \simeq \alpha_2$, if there exists a continuous mapping $H : [0, 1] \times [0, 1] \rightarrow M_{\text{free}}$ such that for all $s, t \in [0, 1]$, $H(s, 0) = \alpha_1(s)$, $H(s, 1) = \alpha_2(s)$, $H(0, t) = \alpha_1(0)$, $H(1, t) = \alpha_1(1)$. The *homotopy class* of a curve α_0 is defined as the set of all curves homotopic to it: $[\alpha_0] := \{\alpha \mid \alpha \simeq \alpha_0\}$.

Intuitively, $[\alpha_0]$ contains all curves that can be continuously deformed into α_0 without crossing obstacles.

Definition 2 (h-Signature of a Curve) The h-signature is used to represent the homotopy class of a curve. It is constructed as follows: for each independent obstacle in the map, select a representative point inside the obstacle and emit a non-intersecting ray from that point. The h-signature of a curve α is defined by the sequence in which the curve crosses these rays from the start to the goal, as illustrated as Fig. 3(a). For two curves with the same start and goal points, if their h-signatures are identical, they belong to the same homotopy class. The addition of h-signatures can be defined based on curve concatenation: if $\alpha = \alpha_1 * \alpha_2$, then $h(\alpha) = h(\alpha_1) * h(\alpha_2)$.

Based on relevant concepts from topology and the mechanical properties of the tether, Lemmas 3 and 4 can be readily obtained.

Lemma 3. In a planar space, the shortest curve within a homotopy class is unique.

Lemma 4. A taut tether configuration corresponds to the shortest curve within a homotopy class.

Lemma 5. Let α_1 and α_2 be the shortest curves in their respective homotopy classes, satisfying the endpoint condition $\alpha_1(1) = \alpha_2(0)$ and being smoothly differentiable at the

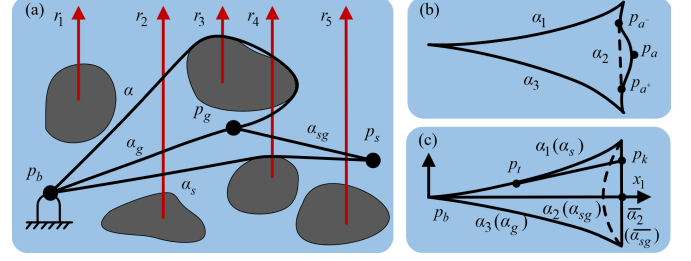


Fig. 3. Illustration of the Definition, Lemma, and Theorem. (a) Illustration related to Definition 2 [14]. $h(\alpha) = r_2 r_3$, $h(\alpha_g) = r_2$, and α, α_g belong to different homotopy classes; $h(\alpha_s) = r_2 r_4 r_5$, $h(\alpha_{sg}) = r_5^- r_4^-$, $h(\alpha_g) = h(\alpha_s) * h(\alpha_{sg})$, hence $\alpha_g \simeq \alpha_s * \alpha_{sg}$. (b) Illustration related to Lemma 9. (c) Illustration related to Lemma 10 and Theorem 11, where α_1, α_2 , and α_3 in Lemma 10 correspond to α_s, α_{sg} , and α_g in Theorem 11, respectively.

connecting point. Then, their concatenation $\alpha = \alpha_1 * \alpha_2$ is also the shortest curve within its homotopy class.

Proof. According to Lemma 4, α_1 and α_2 can be regarded as taut tether configurations. If the connecting point is smooth, the mechanical properties of the tether imply that their concatenation $\alpha = \alpha_1 * \alpha_2$ remains taut and is also the shortest curve within its homotopy class.

Lemma 6. Let α_1, α_2 , and α_3 be the shortest curves in their respective homotopy classes, and suppose that $\alpha_1 * \alpha_2 \simeq \alpha_3$. Then, any pair among α_1, α_2 , and α_3 either shares exactly one common point or has exactly one common subsegment.

Proof. Without loss of generality, consider α_1 and α_2 . If they have two intersection points that do not belong to the same common segment, denote the sub-curves between these points as $\alpha_{1s} \subseteq \alpha_1$ and $\alpha_{2s} \subseteq \alpha_2$. Since α_1 and α_2 are shortest curves in their respective, the sub-curves α_{1s} and α_{2s} are also shortest curves in their respective. Moreover, it is evident that $\alpha_{1s} \simeq \alpha_{2s}$. This contradicts Lemma 3. Therefore, if there are multiple intersection points, they must all lie on the same common segment. Hence, any pair among $\alpha_1, \alpha_2, \alpha_3$ either shares exactly one common point or exactly one common segment.

Definition 7 (Tangent and Normal Vectors). Let $\alpha : [0, 1] \rightarrow \mathbb{R}^2$ be a C^2 smooth curve directed from the start point $\alpha(0)$ to the end point $\alpha(1)$. Denote by $\mathbf{t}(s)$ the unit tangent vector of the curve at $\alpha(s)$, and by $\mathbf{n}(s)$ the unit normal vector obtained by rotating $\mathbf{t}(s)$ clockwise by $\pi/2$.

Definition 8 (Signed Curvature). For any point $\alpha(s)$ on the curve α , $s \in [0, 1]$, the signed curvature is defined as

$$\kappa(s) = \left\langle \frac{d\mathbf{t}}{ds}, \mathbf{n}(s) \right\rangle, \quad (4)$$

where $\langle \cdot, \cdot \rangle$ denotes the vector inner product. The signed curvature describes not only the bending magnitude of the curve but also the direction in which the curve bends relative to the normal vector.

Lemma 9 (Triangular Relationship between Tether Configurations and Path). Let $\alpha_1, \alpha_2, \alpha_3$ be the shortest curves in their respective homotopy classes, with each pair intersecting at exactly one point, and satisfying $\alpha_1 * \alpha_2 \simeq \alpha_3$. Then the signed curvature $\kappa \geq 0$ at all points along α_1, α_2 , and α_3^- .

Proof. Suppose, to the contrary, that there exists a point p_a on α_1, α_2 , or α_3^- such that, within its neighborhood p_a^-, p_a^+ , the

curve segment $\widehat{p_a-p_{a+}}$ contains a point with negative signed curvature, as shown in Fig. 3(b).

Since $\alpha_1 * \alpha_2 \simeq \alpha_3$, the three curves together enclose a Jordan region that contains no obstacles inside. The chord $\overline{p_a-p_{a+}}$ lies entirely within this Jordan region, and its length satisfies $|\overline{p_a-p_{a+}}| < |\widehat{p_a-p_{a+}}|$. Therefore, one can replace $\widehat{p_a-p_{a+}}$ with $\overline{p_a-p_{a+}}$ to construct a shorter curve within the same homotopy class, which contradicts the assumption that α_1 , α_2 , and α_3 are shortest curves.

Therefore, the signed curvature $\kappa \geq 0$ at all points along α_1 , α_2 , and α_3 .

For the slope-tethered robot path planning problem, α_s and α_g are both taut tether configurations. By Lemma 4, they are the shortest curves in their respective homotopy classes. If the path α_{sg} is also the shortest curve in its homotopy class, then according to Lemmas 6 and 9, the non-overlapping portions of α_s , α_g , and α_{sg} always maintain a triangular relationship.

Lemma 10 (Deformation Rule of the Tether during Robot Motion). Let $\alpha_1, \alpha_2, \alpha_3$ be the shortest curves in their respective homotopy classes, with each pair intersecting at exactly one point, and satisfying $\alpha_1 * \alpha_2 \simeq \alpha_3$. Without loss of generality, let $\alpha_1 \cap \alpha_3 = p_b$. Then, for any point p_k on α_2 , either there exists exactly one point p_t on α_1 or α_3 such that the tangent line at p_t passes through p_k , or the line segment $\overline{p_b p_k}$ intersects α_1 and α_3 at exactly one point.

Proof. Without loss of generality, the situation can be illustrated as Fig. 3(c), using the line segment $\overline{\alpha_2}$ to replace α_2 . For the tangent lines or line segments satisfying the condition, their extensions must intersect $\overline{\alpha_2}$. We will prove a stronger statement: every point on $\overline{\alpha_2}$ satisfies the above condition.

Let f_{α_1} and f_{α_3} denote the function form of α_1 and α_3 , respectively. By Lemma 9, $f'_{\alpha_1}(x) \geq 0$, $f'_{\alpha_3}(x) \leq 0$, and $f'_{\alpha_1}(0) \geq f'_{\alpha_3}(0)$.

For a given p_k , define $g(x) = (x_k - x)f'_{\alpha_1}(x) + f_{\alpha_1}(x) - y_k$. If $g(x)$ has a zero x_t , it indicates that the tangent line of α_1 at p_t passes through p_k .

Note that $g(0) = x_k f'_{\alpha_1}(0) - y_k$, $g(x_k) = f_{\alpha_1}(x_k) - y_k$, and $g'(x) = (x_k - x)f''_{\alpha_1}(x)$. For f_{α_1} , $g'(x) \geq 0$ and $g(x_k) > 0$. If $f_{\alpha_1}(0) \leq \frac{y_k}{x_k}$, the mean value theorem guarantees exactly one x_t such that $g(x_t) = 0$. Similarly, for f_{α_3} , $g'(x) \leq 0$ and $g(x_k) < 0$, and if $f_{\alpha_3}(0) \geq \frac{y_k}{x_k}$, there exists exactly one x_t such that $g(x_t) = 0$.

If $f_{\alpha_3}(0) < \frac{y_k}{x_k} < f_{\alpha_1}(0)$, elementary calculus implies that the segment $\overline{p_b p_k}$ intersects α_1 and α_3 at exactly one point.

For the slope-tethered robot path planning problem, if any pair among α_s , α_g , and α_{sg} intersects at exactly one point, then the tether configuration α_k at an intermediate position p_k either coincides with α_s or α_g over an initial segment, or shares exactly one intersection point p_b with them.

C. Theorem

Theorem 11 (Constraint Consistency Theorem). For the tethered robot path planning problem, if α_{sg} is the shortest curve in its homotopy class and the tether configuration at the goal position α_g satisfies the length and maximum deployment angle constraints, then all intermediate tether configurations $[\alpha_s \rightarrow g]$ along the path also satisfy these two constraints.

Proof. Since $\alpha_s * \alpha_{sg} \simeq \alpha_g$ and $\alpha_s, \alpha_{sg}, \alpha_g$ are the shortest curves in their respective homotopy classes, by Lemma 6, any pair among $\alpha_s, \alpha_{sg}, \alpha_g$ either shares exactly one common point or has exactly one common subsegment. Let the tether configuration at the robot's intermediate position p_k be α_k . We first discuss two special cases:

1) p_k lies on the overlapping part of α_{sg} with α_s or α_g : if $p_k \in (\alpha_s \cap \alpha_{sg})$, let $\alpha_s(k) = p_k$, then $\alpha_k(0) = \alpha_s(0) = p_b$, $\alpha_k(1) = \alpha_s(1) = p_k$. By Lemma 3, $\alpha_k \subset \alpha_s$. Similarly, if $p_k \in (\alpha_g \cap \alpha_{sg})$, then $\alpha_k \subset \alpha_g$. Therefore, if the tether configurations at the start and goal positions satisfy the constraints, the intermediate configuration in this case also satisfies the constraints.

2) p_k lies on the non-overlapping part of α_{sg} with α_s or α_g , and α_s, α_g have an overlapping segment: by Lemma 10, for any $p_k \in \alpha_{sg}$, there exists a tangent line α_t from α_s or α_g passing through p_k , with the tangent point located in the non-overlapping part of α_s or α_g . Let the portion of α_s or α_g before the tangent point be α_f . By Lemma 5, $\alpha_k = \alpha_f * \alpha_t$. Since the overlapping portion $(\alpha_s \cap \alpha_g) \subseteq \alpha_k$, the properties of α_k on the overlapping segment are the same as those of α_s and α_g .

After considering these two cases, to prove Theorem 11, it suffices to focus on the scenario in which each pair among α_s, α_{sg} , and α_g shares only one common point. We use $\overline{\alpha_{sg}}$ to replace α_{sg} . Without loss of generality, we assume that $\overline{\alpha_{sg}}$ is perpendicular to the x -axis, and that the starting points of α_s and α_g are located at the origin, as illustrated in Fig. 3(c). Accordingly, in the problem formulation, we have $x \in [0, x_1]$, where $x_{\alpha_s(0)} = x_{\alpha_g(0)} = 0$ and $x_{\alpha_s(1)} = x_{\alpha_g(1)} = x_1$. Since α_k differs from α_s or α_g only along the tangent portion, we always have

$$|f'_{\alpha_k}(x)| \leq |f'_{\alpha_s}(x)| \quad \text{or} \quad |f'_{\alpha_k}(x)| \leq |f'_{\alpha_g}(x)| \quad (5)$$

1) For the length constraint: By the curve length formula

$$L(\alpha) = \int_0^{x_1} \sqrt{1 + f'_{\alpha}(x)^2} dx \quad (6)$$

we have

$$L(\alpha_k) \leq \max(L(\alpha_s), L(\alpha_g)) \quad (7)$$

That is, if the tether configurations at the start and goal positions satisfy the length constraint, the intermediate configurations also satisfy the length constraint, consistent with the result in [21].

2) For the maximum deployment angle constraint:

$$\arctan(|f'_{\alpha_k}(x)|) \leq \max(\arctan(|f'_{\alpha_s}(x)|), \arctan(|f'_{\alpha_g}(x)|)).$$

Assume the angle between v_s and the x -axis is γ_0 , then the maximum deployment angle is

$$\Gamma(\alpha_k) = \arctan(|f'_{\alpha_k}(x)|) + \gamma_0 \leq \max(\Gamma(\alpha_s), \Gamma(\alpha_g)) \quad (8)$$

That is, if the tether configurations at the start and goal positions satisfy the deployment angle constraint, the intermediate configurations also satisfy the maximum deployment angle constraint.

Theorem 12 (Constraint Optimality Theorem). For the tethered robot path planning problem, the globally shortest

path α_{sg}^* satisfying both the tether length constraint and the maximum deployment angle constraint is necessarily the shortest curve in its homotopy class.

Proof. Let α_g be the shortest curve in its homotopy class, uniquely determined by its start point, end point, and h -signature. Since α_{sg}^* is the globally shortest path satisfying the tether constraints, the terminal tether configuration α_g also satisfies the tether constraints. Since $h(\alpha_{sg}^*) = h(\alpha_s^-) * h(\alpha_g)$, the path α_{sg} computed from $h(\alpha_{sg}^*)$ is the shortest path in its homotopy class, and the corresponding terminal tether configuration α_g satisfies the tether constraints. By Theorem 11, α_{sg} is the shortest path in this homotopy class that satisfies the tether constraints. Since the shortest path within a given homotopy class is unique, the globally shortest path satisfying the tether constraints must be the shortest curve in its homotopy class.

The constraint consistency and optimality theorem provide twofold theoretical support for the path planning problem.

Optimality: This theorem ensures that the optimal path α_{sg} must be the shortest curve within a certain homotopy class. Based on this property, the planning problem can be reformulated as follows: **the planner considers all candidate homotopy classes that may contain α_{sg} , computes the shortest path in each candidate homotopy class, and selects the path whose entire set of intermediate tether configurations $[\alpha_s \rightarrow g]$ satisfies both the tether length constraint and the maximum deployment angle constraint. The shortest among these paths is the globally optimal path.**

Consistency: If the robot moves along the shortest path within a certain homotopy class to the goal position, and the tether configuration at the goal α_g satisfies the constraints, then all tether configurations of intermediate points along the path also satisfy the same constraints. Based on this property, considering that under a given initial tether configuration, and when the robot path is the shortest path in its homotopy class, the robot path corresponds one-to-one with the end tether configuration, the planning problem can be further reformulated as follows: **the planner considers all candidate homotopy classes that may contain α_g and selects those in which α_g satisfies both the tether length and maximum deployment angle constraints, then computes the corresponding robot paths. The shortest among these paths is the globally optimal path.**

IV. METHODOLOGY

A. Curve Shortening Algorithm

In a grid map, due to the limitation of grid resolution, paths computed by search algorithms such as BFS or Dijkstra are generally not the shortest curves within their homotopy class. In tethered robot path planning, a curve shortening (CS) algorithm is commonly employed to shrink a given path to the shortest curve within its homotopy class.

The most widely used CS algorithm is that proposed in [14], which has been extensively applied in subsequent tethered robot path planning studies [5], [23]. However, this algorithm does not compute the intermediate anchor positions of the tether and is therefore not suitable for slope-tethered robot

planning problems where the maximum deployment angle constraint must be considered.

To address this, we propose an *h-signature enhanced curve shortening algorithm* (HSE-CS), which can shorten any given curve to the shortest path within its homotopy class under finite resolution while preserving the anchor positions. The detailed procedure is presented in Algorithm 1.

Algorithm 1 HSE-CS

Require: Curve $\alpha = [p_0, p_1, \dots, p_n]$, obstacles M_{obs}

Ensure: Shortest curve α_0 within the same homotopy class

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1:  $\alpha_0 \leftarrow \alpha$ ,  $p_a \leftarrow p_0$ ,  $i \leftarrow 0$ ,  $j \leftarrow n$ 
2: while  $i \neq n$  do
3:   while  $j > i$  do
4:     if  $\text{free}(\overline{p_a p_j})$  and  $(\alpha_0 + \overline{p_a p_j} + \alpha[j :]) \simeq \alpha$  then
5:       if  $j = n$  then
6:          $\alpha_0 \leftarrow \alpha_0 + p_j$ 
7:       else
8:          $p'_a \leftarrow \text{last}(\overline{p_a p_{j+1}} \cap M_{\text{obs}})$ 
9:          $\alpha_0 \leftarrow \alpha_0 + p'_a$ ,  $p_a \leftarrow p'_a$ 
10:      end if
11:       $i := j$ 
12:    else
13:       $j := j - 1$ 
14:    end if
15:  end while
16: end while
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Here, p_a denotes an intermediate anchor point where the tether interacts with obstacles, and $\overline{p_a p_j}$ represents the straight line segment connecting p_a and p_j in the grid map. The function $\text{last}(\overline{p_a p_{j+1}} \cap M_{\text{obs}})$ returns the point immediately after the last intersection of $\overline{p_a p_{j+1}}$ with obstacles M_{obs} , ensuring that $p_a \notin M_{\text{obs}}$.

The HSE-CS algorithm starts from the end of the curve α and iteratively connects the last anchor point in α_0 with straight line segments until the shortened curve preserves the same h -signature as the original curve. The previous straight line segment and the point immediately following its last intersection with obstacles are then taken as the new anchor point. This process continues until the entire curve is traversed, resulting in α_0 as the shortened tether configuration.

B. Planning Algorithm

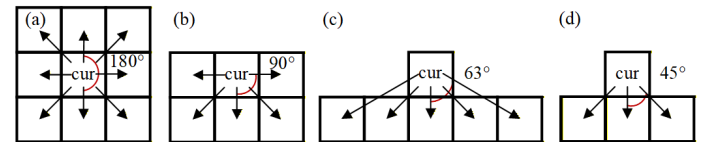


Fig. 4. Receptive field of RFE-Dijkstra. (a) 180°, (b) 90°, (c) 63°, (d) 45°.

To improve the efficiency of slope-tethered robot path planning, this paper proposes a Receptive Field Enhanced Dijkstra algorithm (RFE-Dijkstra), which can reduce the number of candidate paths based on the tether length constraint and the maximum deployment angle constraint. Compared

with the standard Dijkstra algorithm, RFE-Dijkstra introduces improvements in the following two aspects.

First, the RFE-Dijkstra algorithm expands nodes according to the receptive field, where different receptive fields constrain the directions of node expansion. Standard Dijkstra typically adopts an 8-neighborhood assumption, corresponding to a receptive field of 180° (see Fig. 4(a)). However, in slope-tethered robot path planning, the maximum deployment angle (as described in Section III-A) theoretically lies in the range $[0, \pi/2]$. During the search over terminal tether configurations, expanding nodes in the upward direction would violate the maximum deployment angle constraint. Therefore, a receptive field of 90° is sufficient to cover the entire feasible region, corresponding to the lower bound of algorithm performance (see Fig. 4(b)). Due to the discrete nature of the grid map, the minimum granularity of the receptive field is 45° , which corresponds to the upper bound of algorithm performance (see Fig. 4(d)). On this basis, starting from the 45° receptive field, intermediate receptive fields between 45° and 90° can be constructed by increasing the number of lateral expansion nodes, such as 63° (see Fig. 4(c)).

Second, the termination condition of RFE-Dijkstra is not to expand to the goal position, but to limit the search range according to the tether length constraint. Although the length of the shortest path searched by RFE-Dijkstra may exceed the length of the taut tether configuration, the deviation is bounded. Given an inflation factor d and a maximum length $\bar{l}$, the maximum cumulative path length of expanded nodes is $d\bar{l}$. Moreover, whenever the goal position is reached during expansion, the h-signature of the path from the start to the goal is computed to determine its homotopy class. In RFE-Dijkstra, each candidate homotopy class corresponding to the tether configuration at the goal position stores only a single shortest path, which removes duplicate homotopy classes before applying the HSE-CS algorithm.

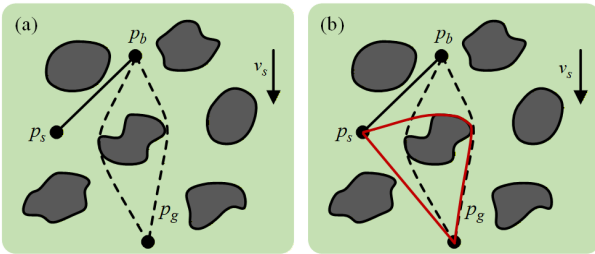


Fig. 5. Computation process of the slope-tethered robot path planning algorithm. (a) Search for candidate terminal tether configurations from p_b to p_g using the RFE-Dijkstra algorithm, (b) Computation of the robot path corresponding to each terminal tether configuration.

Based on the conclusions in Section III, the slope-tethered robot path planning algorithm consists of three steps, as shown in Algorithm 2. First, the RFE-Dijkstra algorithm is used to compute all candidate paths $\{\alpha_{gi}\}$ from p_b to p_g , as shown in Fig. 5(a). Second, the HSE-CS algorithm shortens all candidate paths to obtain the taut tether configurations $\{\alpha_{gi}\}$. Finally, the robot paths $\{\alpha_{sgi}\}$ corresponding to the feasible $\{\alpha_{gi}\}$ are computed, as illustrated in Fig. 5(b). Here, $h\text{-Dijkstra}(p_s, p_g, h(\alpha_{sg}))$ denotes searching for the shortest

path from p_s to p_g within the homotopy class specified by $h(\alpha_{sg})$. Compared with the standard Dijkstra algorithm, the only difference is that, during node expansion, it is required to verify whether the h-signature of the path traced back from the newly expanded node to p_s is a subsequence of $h(\alpha_{sg})$. The shortest path among them is selected as the optimal path α_{sg} .

Algorithm 2 Tethered Robot Path Planning Algorithm

Require: Grid map M , anchor point p_b , current position p_s , current tether configuration α_s , goal position p_g , maximum length $\bar{l}$, maximum deployment angle $\bar{\gamma}$

Ensure: Robot path α_{sg}

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1:  $\{\alpha_{gi}\}_{i=1 \sim n} \leftarrow \text{RFE-Dijkstra}(M, p_b, p_g)$ 
2: for  $\alpha_{gi}$  in  $\{\alpha_{gi}\}_{i=1 \sim n}$  do
3:    $\alpha_{gi} \leftarrow \text{HSE-CS}(\alpha_{gi})$ 
4:   if  $L(\alpha_{gi}) \leq \bar{l}$ ,  $\Gamma(\alpha_{gi}) \leq \bar{\gamma}$  then
5:      $h(\alpha_{sg}) \leftarrow h(\alpha_s^-) * h(\alpha_{gi})$ 
6:      $\alpha_{sg} \leftarrow h\text{-Dijkstra}(p_s, p_g, h(\alpha_{sg}))$ 
7:      $\alpha_{sg} \leftarrow \text{HSE-CS}(\alpha_{sg})$ 
8:   end if
9: end for
10:  $\alpha_{sg} \leftarrow \arg \min_{\{\alpha_{sgi}\}_{i=1 \sim n}} L(\alpha_{sgi})$ 

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V. EXPERIMENTS

A. Experimental Setup

To systematically evaluate the effectiveness of the proposed tethered robot path planning algorithm on slopes, we conducted multiple simulation experiments in map environments with varying numbers of obstacles. The map dimensions are set as $H = W = 200$. In each map, 20 feasible initial tether configurations and goal positions are randomly generated. For the tether constraints, the maximum tether length is set as $\bar{l} = 150$, and the maximum deployment angle is $\bar{\gamma} = 0.78$ radians (45°).

To validate the effectiveness of the proposed planning algorithm and the role of constraint consistency and optimality, three baseline algorithms are considered:

1) **Selecting the Shortest Path (SSP)**: It directly uses Dijkstra to plan all candidate paths from p_s to p_g , and then evaluates whether each candidate path satisfies the tether constraints. Different from our method, SSP relies only on the constraint optimality theorem and evaluates tether constraints at every point along the candidate path.

2) **Planning the Shortest Path (PSP)** [20]: It directly uses Dijkstra to plan the path from p_s to p_g , checking the tether state for all expanded nodes along the way (i.e., search and constraint evaluation are coupled).

3) **Preprocessing the Map (PM)** [14]: It first computes the reachable space of the robot relative to the anchor p_b , removes all positions that violate tether constraints, and then performs a standard tether-free path planning within the remaining reachable space.

B. Experimental Results

The experiments were conducted on a computing platform equipped with an AMD Ryzen 9 7900X processor. Since PM

is equivalent to tether-free path planning during the planning stage, we report its computation time as the time required for a single map preprocessing run. All results are averaged over 20 times runs.

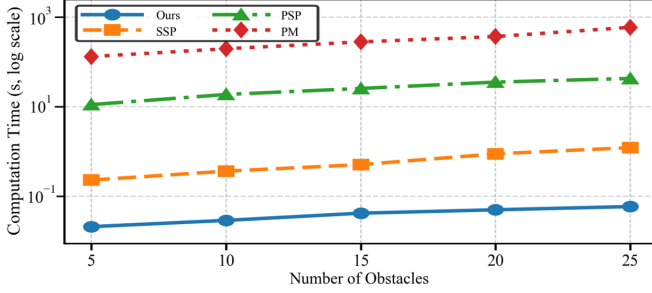


Fig. 6. Comparison of computation times of different algorithms under varying numbers of obstacles.

To highlight the computational efficiency of the proposed algorithm, we compared the average solving times, as shown in Fig. 6, the vertical axis is in logarithmic scale. When the number of obstacles is 5, the computation times of our algorithm, SSP, PSP, and PM are 21 ms, 230 ms, 11.1 s, and 131 s, respectively. When the number of obstacles increases to 25, the computation times are 59.5 ms, 1,220 ms, 52.6 s, and 598 s, respectively. The experimental results indicate that the average computation time of the proposed algorithm is approximately 1/15 of SSP, 1/600 of PSP, and 1/7000 of PM. We attribute the significant differences in computation time to two main factors:

First, the constraint optimality theorem allows the algorithm to avoid searching for the shortest path satisfying the constraints among all possible paths from the start to the goal (as in PSP). Instead, it can directly select the shortest path satisfying the constraints from the shortest paths of candidate homotopy classes (as in SSP). In a grid map, the number of paths within a homotopy class is enormous, resulting in SSP having a computation time approximately 40 times longer than PSP. Second, the constraint consistency theorem enables the algorithm to determine whether the entire path satisfies the constraints solely based on the tether configuration at the goal position (as in ours). If a path α_{sg} contains n waypoints, our method only requires a single call to the *HSE-CS* algorithm, whereas SSP needs $n - 1$ calls to *HSE-CS*. Consequently, our computation time is approximately 15 times faster than SSP.

In contrast, PM must compute the tether constraints for all homotopy classes at all grid points, resulting in a computational cost far exceeding the other methods. PM is thus only suitable for scenarios where both the environment and the anchor point are fixed.

We conducted a detailed comparison of the planning results from the four algorithms, as shown in Fig. 7. Notably, thanks to the *HSE-CS* algorithm, both our proposed method and the SSP algorithm are able to produce the optimal path. In contrast, PSP and PM, constrained by the use of standard grid search methods such as BFS, Dijkstra, or A*, do not necessarily yield globally optimal paths. Only by applying the *HSE-CS* algorithm based on the constraint consistency

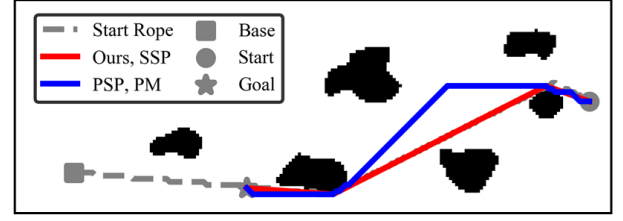


Fig. 7. Comparison of outputs from different algorithms. The results show that our method and SSP both achieve the optimal path with $l = 82.9$, while PSP and PM produce paths that are clearly suboptimal, $l = 88.0$.

theorem can these paths potentially be shortened to achieve global optimality.

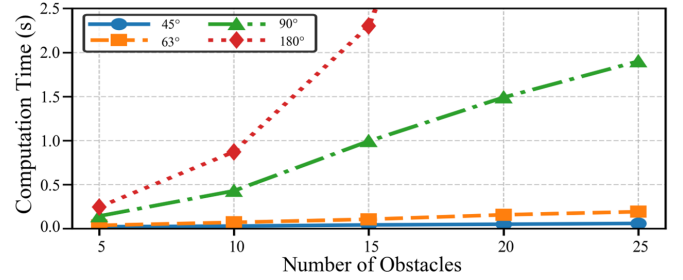


Fig. 8. Impact of receptive field on the planning time of RFE-Dijkstra under different numbers of obstacles. The average computation time with a 45° receptive field is only 1/94 of that with a 180° receptive field. When the number of obstacles increases from 5 to 25, the computation time with a 45° receptive field increases by 2.8 times, while for a 180° receptive field it increases by 54 times.

We compared the computation time of the RFE-Dijkstra algorithm under different receptive fields, as shown in Fig. 8. The results indicate that a smaller receptive field leads to higher search efficiency. When the number of obstacles exceeds 15, the RFE-Dijkstra algorithm (with receptive fields of 45° , 63° , and 90°) significantly outperforms the standard Dijkstra algorithm (with a receptive field of 180°) in terms of computation time. Moreover, the upper bound of RFE-Dijkstra performance (with a 45° receptive field) is substantially better than its lower bound (with a 90° receptive field), and the performance of intermediate receptive fields (with a 63° receptive field) is closer to the upper bound. In addition, when the receptive field is no smaller than the maximum deployment angle constraint, the shortest paths computed under different receptive fields are identical. However, when the receptive field is smaller than the maximum deployment angle constraint, some of the obtained solutions become suboptimal. Therefore, the receptive field of the RFE-Dijkstra algorithm should be no smaller than the maximum deployment angle constraint and, in practice, should be chosen to be as close to it as possible.

Finally, we validated the proposed algorithm in a scenario containing 25 obstacles, under various anchor points as well as start and goal positions, as shown in Fig. 9. The results demonstrate that the intermediate tether anchor points are accurately positioned and satisfy the physical constraints, and that the triangular relationship between the tether configuration and the robot path is clearly preserved.

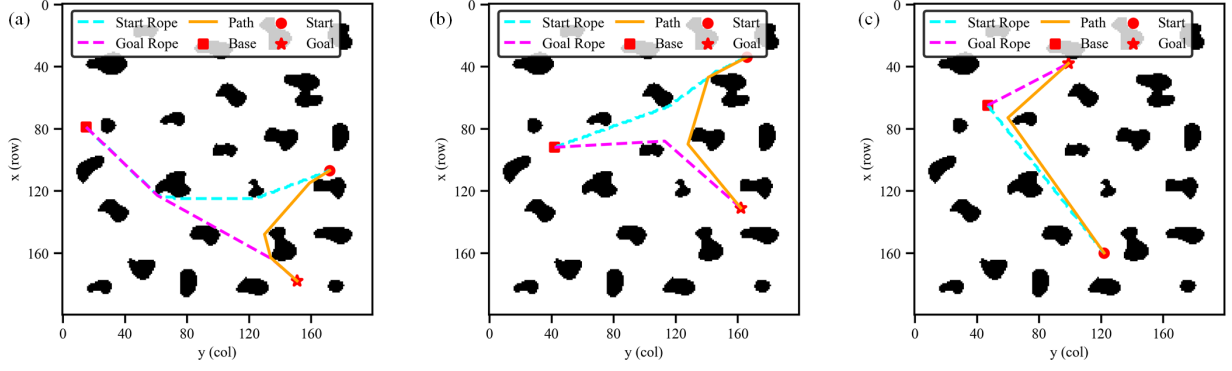


Fig. 9. Planning results of our algorithm on a 200×200 map with 25 obstacles, $\bar{l} = 200$, $\bar{\gamma} = 1.05$. (a) $p_b = (79, 15)$, $p_s = (107, 172)$, $p_g = (178, 151)$, $L(\alpha_{sg}) = 97.4$, $l_{max} = 178$, $\gamma_{max} = 0.76$. (b) $p_b = (92, 42)$, $p_s = (34, 166)$, $p_g = (131, 162)$, $L(\alpha_{sg}) = 126$, $l_{max} = 138$, $\gamma_{max} = 0.72$. (c) $p_b = (65, 47)$, $p_s = (160, 122)$, $p_g = (38, 99)$, $L(\alpha_{sg}) = 159$, $l_{max} = 121$, $\gamma_{max} = 0.90$.

VI. CONCLUSION

This paper investigates the path planning problem for tethered robots in extreme environments. The proposed CCO theorem decouples the evaluation of constraints, such as tether length and maximum deployment angle, from the search process. The RFE-Dijkstra algorithm achieves an approximately 54 times speedup in generating candidate tether configurations. Benefiting from these advantages, the proposed method attains an overall speedup of nearly 600 times compared to conventional search-based approaches.

Nevertheless, this work assumes an offline planning setting with prior knowledge of obstacle locations and models the terrain as a simplified planar slope, which limit its applicability in dynamic or highly complex 3D environments. Extending the framework to online and more realistic settings is an important direction for future work.

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